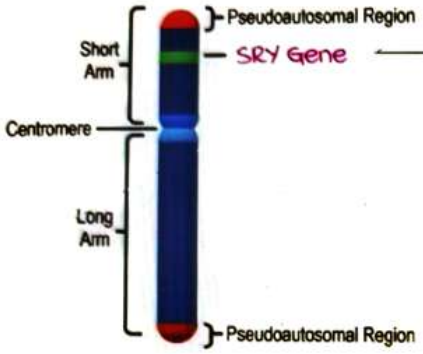
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----- Active space -----

MALE REPRODUCTION

Sex-determining region of Y chromosome (SRY):



Testis-determining gene produces (Transcription factors)

Acts on

- Leydig cells → Testosterone → Wolffian system stimulation → male organs development
- Sertoli cells → Anti-mullerian hormone → Inhibits female organs development

- Absence of SRY → Female.

Puberty

00:00:30

Pre-pubertal:

Continuous release of GnRH (−) → LH, FSH

Maintained by GABA, melatonin

At puberty:

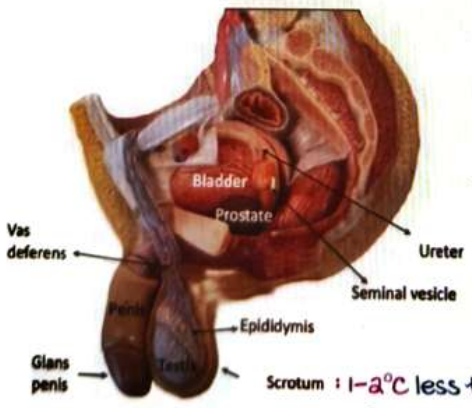
Adequate fat stores → ↑ leptin (Permissive role in puberty) → ↑↑ Kiss peptin

GPR54 in GnRH neuron → Pulsatile release of GnRH

When GnRH ↓ → LH, FSH (+)

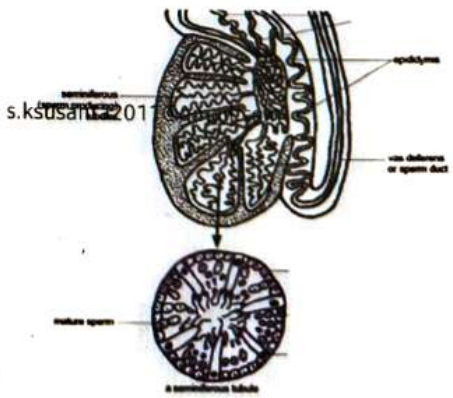
Male Reproduction

00:10:06



Anatomy

Scrotum: 1–2°C less than body temp to favour spermatogenesis



Testis

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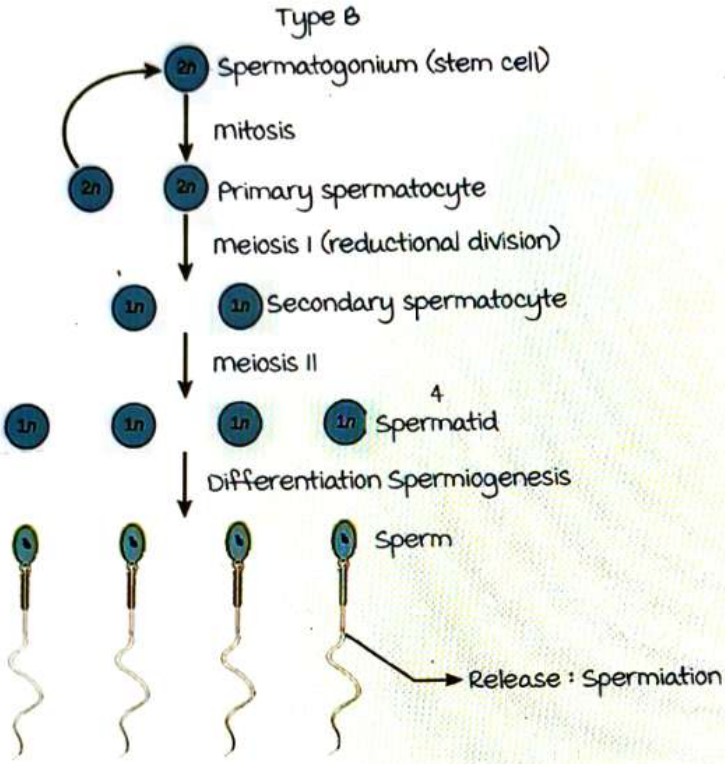
SPERMATOGENESIS

----- Active space -----

Location : Seminiferous tubules.

Duration : 74 days.

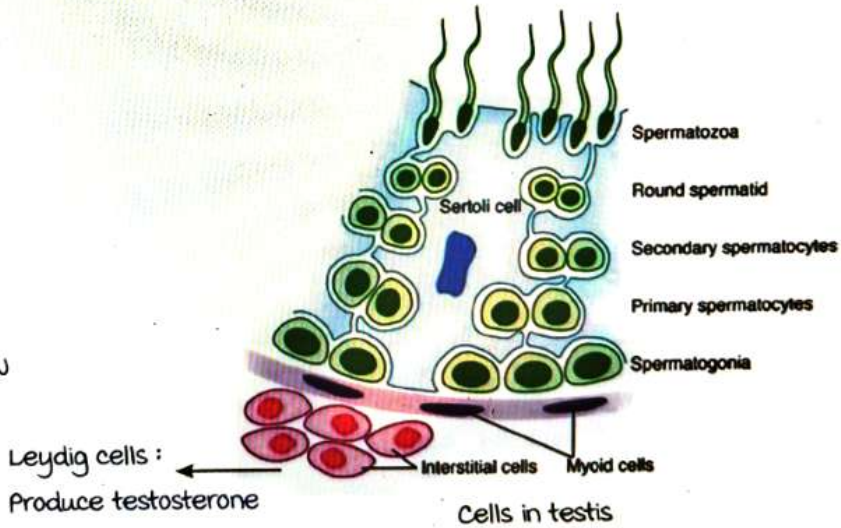
Process :



CELLS IN TESTIS

Functions of Sertoli Cells :

- Production of anti-mullerian hormone (AMH) : Regression of female organ development.
- Blood-testis barrier : Tight junctions b/w 2 adjacent sertoli cells.
- "Nurse cells of testis" : Provide nourishment & protection to sperms.
- Production of Androgen-binding protein (ABP) : Binds testosterone.
- Aromatase : Testosterone → Estrogen.
- Inhibin : Inhibits FSH.
- Phagocytic function : Ingestion of dead sperms.



Cells in testis

Feedback

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----- Active space -----

SEMEN

Contents :

Sperm + Fluid —Produced by→
- Seminal vesicle (60 %).
- Prostate (30 %).
- Vas deferens (10 %).



Semen

pH : 7.2–7.4.

Sperm count : 90–120 million/mL.

Sperm :

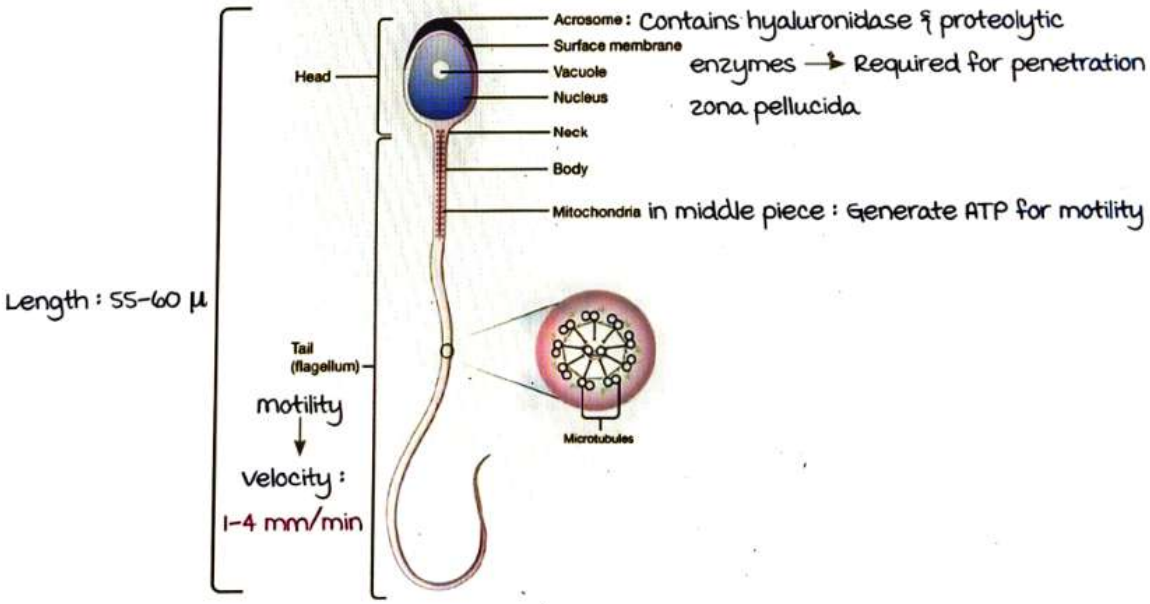
Life span : 1–2 days following ejaculation.

Sperm maturation :

- Acquirement of motility in epididymis :
  - ↑ forward motility by Ca^{a+} channel catsper.
- maturation of acrosome.

Note :

Vaginal secretion = ↑ sperm motility



Sperm capacitation :

- After ejaculation into female genital tract.
- Acrosome reaction : Release of enzymes for penetration of zona pellucida → Fertilization.

Feedback

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Fluid component :

----- Active space -----

Produced by

Seminal Vesicle (60 %) :

- Fructose : Energy source (GLUT-5 mediated).
- Ascorbic acid/Vit C : Antioxidant.
- Prostaglandin : makes cervical mucus penetrable by sperm.
- Phosphorylcholine : Detected via Florence test (Test for seminal stain in rape cases).

Prostate (30 %) :

- Fibrinolysin : Liquefaction of semen in 15–30 min.
- Acid phosphatase.
- Spermine : Detected via Barberio's test.

Vas deferens (10 %) :

- Bicarbonate.
- Phosphate.

Buffers : Bicarbonate, phosphate.

Male Sexual Act

00:32:51

1. Erection of penis :

Facilitated by :

1. Parasympathetic (S2, S3, S4 roots) : mediated by acetylcholine.
2. Non-adrenergic non-cholinergic (NANC) system :
   - Nitric oxide → Dilatation of arterioles → Compression of outflow veins → Engorgement & erection (Acts via cGMP).

Applied Aspect :

- cGMP degraded by phosphodiesterase.
- Phosphodiesterase inhibitor (Sildenafil) → ↑ cGMP → ↑ NO (nitric oxide) → Erection.

2. Ejaculation :

- Sympathetic (T12–L2 nerve roots).
- Two steps : Emission → Ejaculation.
  (movement of semen into urethra) (Release of semen from urethra)

3. Resolution (Detumescence) :

Endothelin & Noradrenaline → Constriction of arterioles → Resolution.

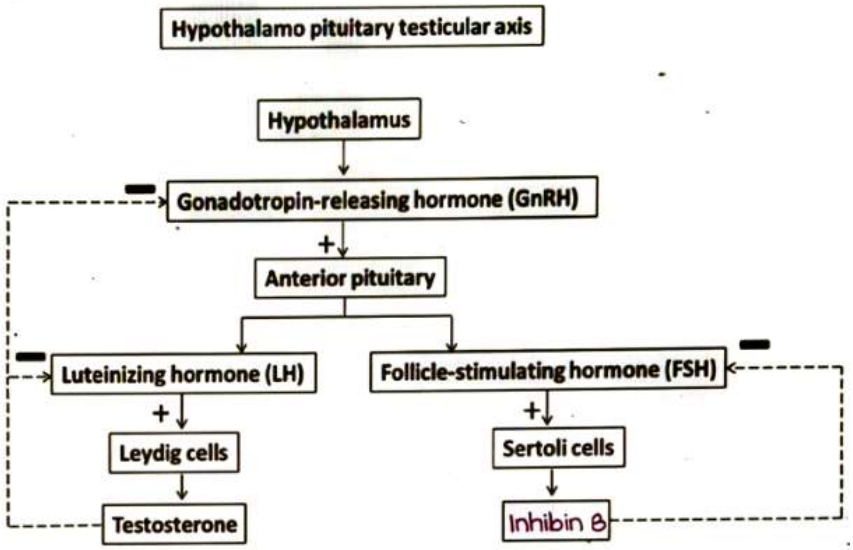
Feedback

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Regulation of Male Reproduction

00:39:30

Two cell two gonadotrophin model :



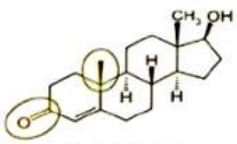
Note : Prolonged administration of testosterone causes azoospermia (D/t negative feedback on LH).

Regulation of spermatogenesis :

- Testosterone.
- FSH.
- LH.
- Growth Hormone (Early division of spermatogonia).

Male Sex Hormones

00:42:36



Testosterone —5α reductase→ Di-hydro testosterone (DHT) (more potent)

- Stimulates wolffian duct → Formation of male internal genital organs.
- Stimulates erythropoietin production → ↑ RBC.
- Protein anabolism → ↑ muscle mass in males.
- Stimulates bone growth.
- Initiation & maintenance of spermatogenesis.
- Stimulate synthesis of ABP from Sertoli cells.
- Behavioral effects : ↑ libido in men.

- Responsible for puberty-associated changes.
- masculinization of external genitalia.
- Development of prostate gland (5α reductase inhibitors finasteride used in Rx of benign prostatic hyperplasia).
- Growth of facial hair.
- Responsible for acne.
- Temporal recession of hairline in males (male pattern baldness).

Feedback